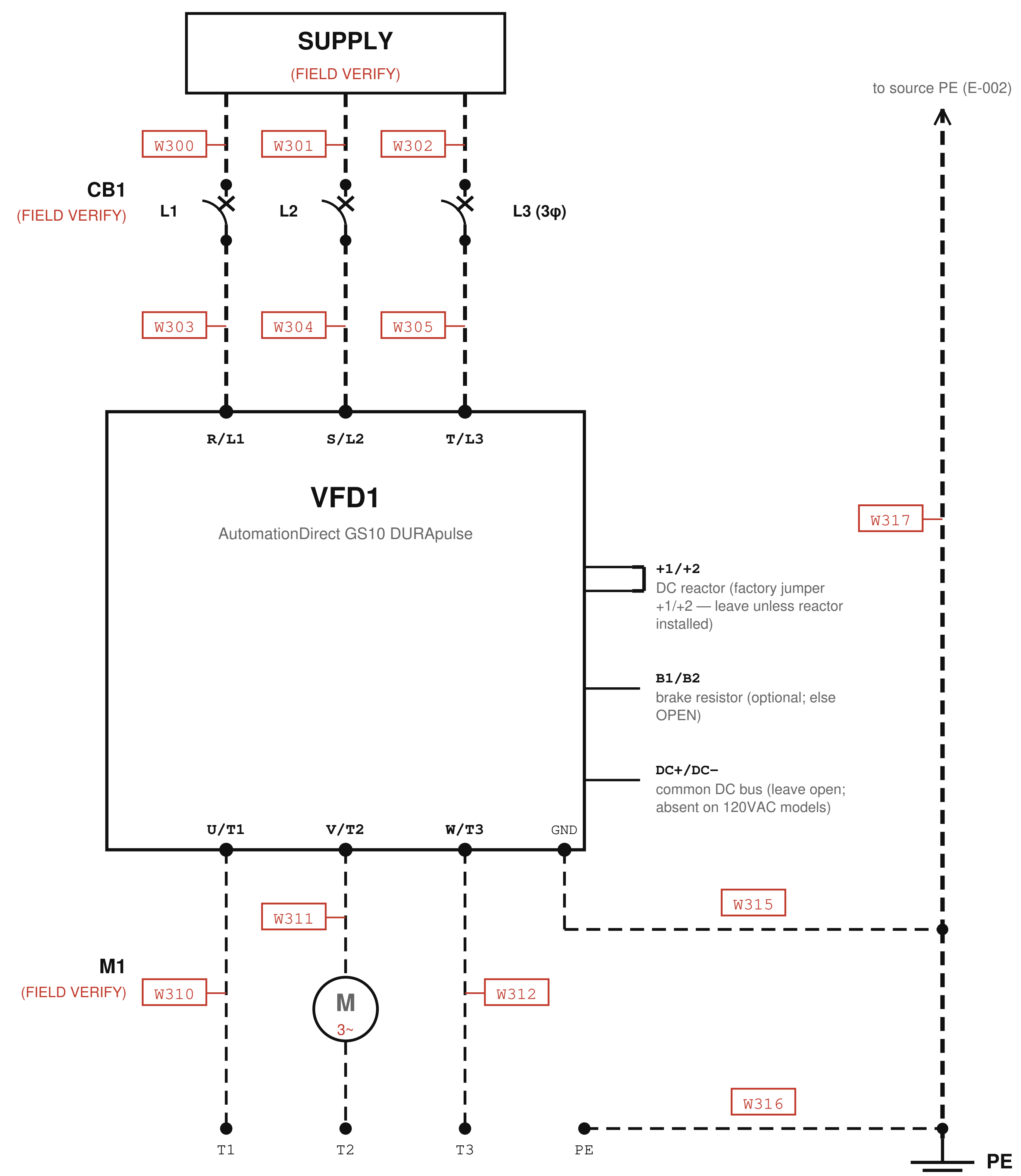
E-003 VFD POWER

Conv_Simple / CV-101 (garage conveyor) — supply → CB1 → VFD1 (GS10) → M1 · every conductor FIELD VERIFY



CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W300	SUPPLY (E-002)	CB1.1	L1 supply	power_line	field_verify	
W301	SUPPLY (E-002)	CB1.3	L2/N supply	power_line	field_verify	
W302	SUPPLY (E-002)	CB1.5	L3 supply (if 3φ)	power_line	field_verify	phase count unknown
W303	CB1.2	VFD1.R/L1	L1 to drive	power_line	field_verify	
W304	CB1.4	VFD1.S/L2	L2 to drive	power_line	field_verify	
W305	CB1.6	VFD1.T/L3	L3 to drive (if 3φ)	power_line	field_verify	phase count unknown
W310	VFD1.U/T1	M1.T1	motor phase U	motor_lead	field_verify	
W311	VFD1.V/T2	M1.T2	motor phase V	motor_lead	field_verify	
W312	VFD1.W/T3	M1.T3	motor phase W	motor_lead	field_verify	
W315	VFD1.GND	PE bus	drive PE (≤0.1Ω)	ground	field_verify	
W316	M1.PE	PE bus	motor frame PE	ground	field_verify	
W317	PE bus	SUPPLY (E-002)	PE to source	ground	field_verify	

Bench supply voltage & phase count, GS10 exact model/frame, breaker rating, wire gauge: NOT DOCUMENTED — every conductor FIELD VERIFY. P00.01 = 1.60 A (2026-05-20 export) is the only sizing clue. If 1φ model, input = R/L1, S/L2 only (GS10_UM L1971).
No 3-phase motor contactor observed in the 2026-07-11 photos. The device labeled 'MLC' is a Schneider TeSys CA3KN22BD CONTROL RELAY (drive-enable interlock) shown on E-006 — not a motor contactor. Separate power contactor = FIELD VERIFY (OI-21).

SAFETY

- Never start/stop via input power (L1811-1813); CB1 is branch protection only, not a run/stop device.
- LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
- Monitored e-stop is NOT a safety stop — hardwire S0 to remove drive power per NFPA 79 / EN 60204-1.
- PE resistance ≤0.1Ω (L1760-61); shield/conduit bonded both ends (L1792-95); RFI jumper in/out FIELD VERIFY (L1693-1718, OI-17).

NOTES

- CB1 REQUIRED per GS10_UM L1758-1759 — type/rating unknown (OI-15).
- M1: swap any two motor leads to reverse — GS10_UM L1773-1776.
- VFD control source = Modbus — see E-007. No control wiring on this sheet.

SOURCES:

plc/GS10_UM.txt (terminals L1971-1986; topology L1750-1813) · plc/GS10_Integration_Guide.md (Q1 phase-5 test)
plc/Prog_init_ConvSimple_v2.1.st (vfd_run_permit) · plc/MIRA_PLC_WorkInstruction_v3.pdf (LOTO §2)
photos review/photos/wire_1..4.jpg (2026-07-11, actual devices)

LEGEND

VERIFIED (solid)
FIELD VERIFY (dashed)
Wxxx proposed wire number (red = unverified)

VFD POWER Conv_Simple / CV-101 (garage conveyor)	SHEET E-003
Drawn: MIRA / FactoryLM terminals per GS10 UM 1st Ed Rev B;	REV A 3 of 9 2026-07-10